

AI-Powered Fashion Recommendation & Virtual Try-On System

RARITONE SUMMER INTERNSHIP 2026



Introduction

- The fashion industry is increasingly adopting Artificial Intelligence (AI) and Computer Vision technologies to provide personalized and interactive shopping experiences. Traditional online shopping often makes it difficult for customers to visualize how clothing will look on them before purchasing.
- To address this challenge, this project explores AI-powered solutions such as body detection, pose estimation, cloth segmentation, virtual try-on systems, and fashion recommendation models that can improve user experience and support smarter fashion decisions.

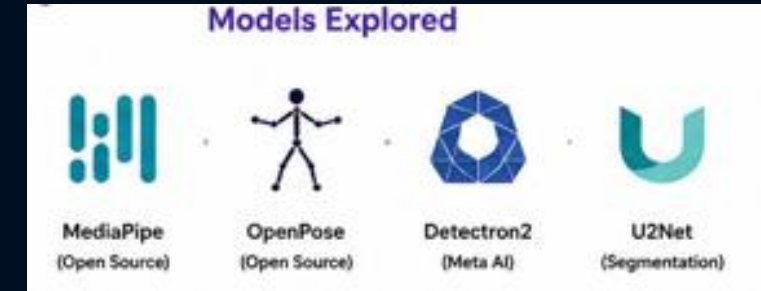
Research Areas Explored Throughout the Internship

- Computer Vision
- **Fashion AI**
- Virtual Try-On
- Recommendation Systems
- **AR/VR Technologies**
- Datasets

Models Explored

Purpose

To evaluate different AI models used for body analysis, garment segmentation, virtual try-on, and fashion recommendations.



Category	Models
Pose Detection	MediaPipe, OpenPose
Segmentation	U2Net, Detectron2
Virtual Try-On	HR-VITON, VITON-HD
Recommendation	Content-Based, Collaborative, Hybrid

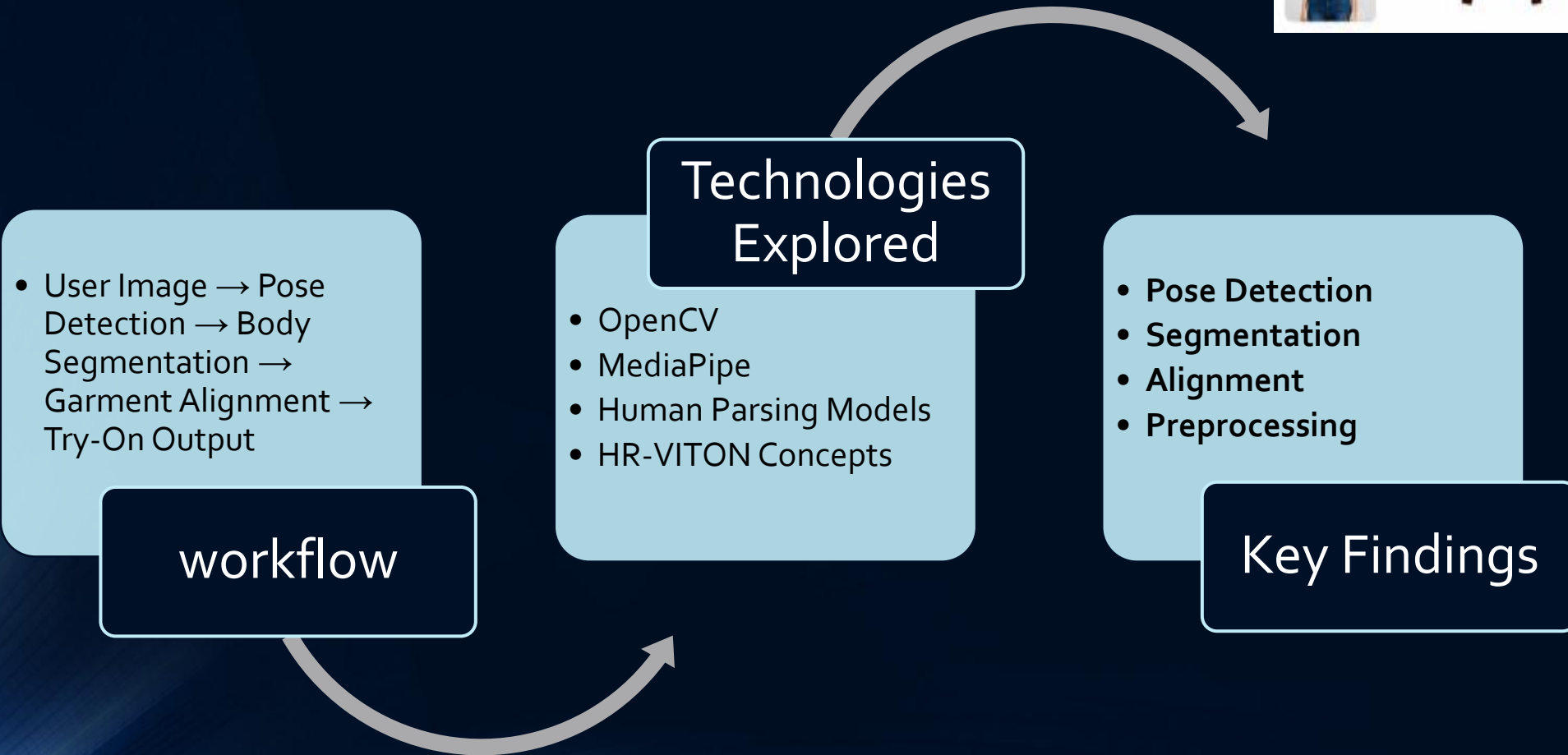
Datasets Collected

- **Fashion & Virtual Try-On Datasets**
- **DeepFashion** – Clothing attributes and fashion images
- **DeepFashion2** – Garment detection and segmentation
- **VITON-HD** – Virtual try-on research and garment fitting
 - **Human Analysis Datasets**
 - Human Pose Datasets
 - Body Segmentation Datasets

Usage

- Pose Detection
- Cloth Segmentation
- Virtual Try-On Research
- Model Testing & Evaluation

2D Virtual Try-On Findings



3D Virtual Try-On Findings

RESEARCH AREAS

- 3D Avatar Generation
- Body Mesh Reconstruction
- Cloth Simulation
- Motion Tracking

TOOLS EXPLORED

- Blender
- Unity
- Three.js
- SMPL Models

Key Findings

- Realistic Visualization
- Interactive Experience
- Real-Time Fitting
- Higher Computational Requirements



AI Fashion Recommendation System

Recommendation Approaches

- Content-Based Filtering
- Collaborative Filtering
- Hybrid Recommendation System

Recommendation Factors

- User Preferences
- Body Type
- Fashion Trends
- Shopping History

BENEFITS

- Personalized Suggestions
- Better User Experience
- Improved Product Discovery
- Increased Customer Engagement

Complete AI Workflow

User Upload



Body Detection



Pose Estimation



Body Segmentation



Measurement Analysis



Garment Mapping



Virtual Try-On



Fashion Recommendation



Final Output

Outcome

- Personalized Fashion Experience
- Improved Try-On Accuracy
- Smart Product Recommendations



Challenges Faced & Solutions

Key Outcome

- Improved understanding of AI fashion workflows
- Better analysis of virtual try-on technologies

Challenges	Solutions
Pose Variations	Improved Landmark Detection
Clothing Alignment	Better Garment Mapping
Segmentation Accuracy	Advanced Segmentation Models
Dataset Limitations	Multiple Dataset Research
Processing Complexity	Workflow Optimization

Future Improvements & Suggestions for Raritone

Future Improvements

- Real-Time Virtual Try-On
- Enhanced 3D Cloth Simulation
- Improved Body Measurement Accuracy
- Faster AI Processing
- Mobile Application Integration

Suggestions for Raritone

- Develop an AI Personal Stylist
- Introduce AR-Based Fashion Try-On
- Support Multiple Body Types
- Expand Fashion Dataset Collection
- Implement Advanced Recommendation Systems

Key Learnings

Technical Learnings

- Computer Vision Fundamentals
- Human Pose Estimation
- Cloth Segmentation Techniques
- Virtual Try-On Workflows
- AI Recommendation Systems
- Dataset Analysis & Research

Professional Learnings

- Team Collaboration
- Research & Documentation
- Problem Solving
- Project Coordination
- Presentation Skills

OUTCOME

Gained practical knowledge of AI-powered fashion technologies and their applications in modern e-commerce systems.

Conclusion

- Explored AI applications in fashion technology.
- Studied 2D and 3D Virtual Try-On systems.
- Analyzed AI models and datasets.
- Investigated recommendation system approaches.
- Developed an understanding of end-to-end AI fashion workflows.

Thank You

Raritone Summer Internship 2026

*Thank you for the opportunity to learn, research,
and contribute throughout this internship.*